BAO (BILLY) DO

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SUMMARY

Master student in National Yunlin University of Science and Technology (YunTech), International Graduate School of Artificial Intelligence, holding a Bachelor's degree in Information Technology from the University of Information Technology (UIT, VNU-HCM, Vietnam). Drawing on a strong background in data analysis, research, and academia, my work focuses on applied artificial intelligence, deep learning, and data-driven insights, spanning topics such as social commerce and public sentiment analysis. This research appears in international journals including Acta Psychologica as well as proceedings of conferences such as ICIS (TREOS), PACIS, BIGS. Alongside my research, I have hands-on experience as an AI & Data Science research and teaching specialist, designing courses, developing RAG-based LLM applications, and applying machine learning solutions to real-world problems.

EDUCATION

Master of Science, Artificial Intelligence **Sep 2025 - now**
National Yunlin University of Science and Technology (Yuntech), Douliu, Yunlin County, Taiwan, R.O.C.

Bachelor of Information Technology **June 2024**
University of Information Technology, Vietnam National University HCMC, Ho Chi Minh City, Vietnam.

Concentrations: Deep Learning, Data Analysis.

Graduation thesis: Image Captioning for Healthcare domain in Vietnamese VieCap4H dataset.

Advisors: M.Sc Vo Duy Nguyen, Dr. Nguyen Tan Tran Minh Khang.

SKILL SETS

Data Analysis: Utilize statistical methods and exploratory analysis to uncover significant insights within the data.

Data Crawling: Collect data from social media/ e-commerce platforms.

Data Processing & Wrangling: Clean, transform and prepare data for analysis. Proficient in Python libraries like Pandas, Numpy, Matplotlib, Seaborn, along with SQL queries, Common Table Expression (CTE) and Window Functions.

Data Visualization: Create impactful charts, graphs and dashboards using tools like PowerBI and Tableau to effectively communicate data insights.

Database Management: Skilled in working with relational databases (MS SQL Server, MySQL) and cloud-based solutions like Big Query.

Machine Learning Fundamentals: Possess understanding of unsupervised learning (k-means, PCA) and supervised Learning (linear/logistic regression, KNN, SVM).

Programming: Python, Javascript, Pyspark, Latex.

Research Tools: PowerBI, Tableau, Python, Visual Studio Code, Google Colaboratory.

WORKING EXPERIENCE

Computer Science School – University of Science – Vietnam National University HCMC

July 2024 – May 2025

AI – Data Science Research and Teaching Specialist

Improving and teaching courses related to Exploratory Data Analysis, Data Mining.

- Instructing trainees how to visualize data with Python by using two libraries Matplotlib and Seaborn to analyze and explore data and discover valuable insights from data. (This course details: https://csc.edu.vn/data-science-machine-learning/data-manipulation-and-visualization-with-python-version-2024_287)
- Assisting trainers in creating visually appealing Tableau reports and dashboards to effectively communicate data analysis messages to viewers. (This course details: https://csc.edu.vn/data-analysis/truc-quan-hoa-du-lieu-data-visualization_305)
- Improving the course "Artificial Intelligence and Applications for Business" – This course helps trainees better understand the relationship between data, artificial intelligence, machine learning and how to apply modern Generative AI tools in the workplace. (This course details: https://csc.edu.vn/data-science-machine-learning/artificial-intelligence-and-applications-for-business_294)

Participating in researching and building AI Chatbot to answer trainees' questions.

- Building AI chatbot to help answer students' questions based on internal documents using RAG architecture.
- Technologies:
    - Retrieval Augmented Generation (RAG) technique.
    - Vector Database: ChromaDB.
    - Embedding model: Sentence Transformer (vietnamese-sbert).
    - LLMs model: Gemini 1.5 Flash (from Google AI Studio).

Saigon A.I.

June 2023 – September 2023

Data Analyst

Learning experience about Exploratory Data Analysis, Data Mining, Storytelling.

- Learning WEKA courses – Machine Learning and Data Mining courses from University of Waikato.
- Using statistical method (T – test, Z – test, Chi – squared) to evaluate performance of which hypotheses are better.
- Using Python to visualize charts to find the insights and make a report.

Projects:

1. **Exploratory Data Analysis – Wine quality dataset**
    - Preprocessing dataset by using Python to handle missing data.

- Creating hypotheses testing and pick one test (T – Test, Z – Test, Chi – Squared), based on p-value to evaluate performance which are better.
- Using Python to visualize charts to find the insights and make a report.

2. Customer Behavior Analysis – ForMotiv dataset

- Cleansing ForMotiv dataset – Insurance dataset by removing missing and noise data.
- Using Python to create charts to answer company inquires and make a report.

Emage Development Co., Ltd **October 2022 – December 2022**
AI Engineer – Internship

Learning experience about Deep Learning, especially Computer Vision.

- Researching the Few-shot Learning and lightweight models such as MobileNetV2, MobileNetV3 to solve problems with small amounts of data.
- Researching and building VietOCR model-based to solve Optical Character Recognition (OCR) task.

CONFERENCE PROCEEDINGS

[1] **B. G. Do**, D. C. Bui, N. D. Vo and K. Nguyen, "A Multi-scale Approach for Vietnamese Image Captioning in Healthcare Domain," *2022 9th NAFOSTED Conference on Information and Computer Science (NICS), Ho Chi Minh City, Vietnam, 2022.*

[2] **Bao Do,** Tuan-Anh Ngo, Sathwik Krishtipati, Thi Tran, Sumantra Sarkar. "My Data Matters: Examining Public Sentiments and Reactions Facing Data Breach Incidents". *Bright Internet Global Symposium (BIGS), Pre-International Conference on Information Systems (ICIS) Virtual Workshop, Hyderabad, India, 2023.*

[3] Tran, Thi; **Do, Bao Gia**; Ngo, Anh; Krishtipati, Sathwik; Dang, Nhat Anh; and Sarkar, Sumantra, "The Impacts of Layoffs Announcement on Cybersecurity Breaches" (2024). *PACIS 2024 Proceedings. 10. [CORE-A, SCOPUS]*

[4] **Do, Bao**; Nguyen, Huan Hong; Tran, Thi; and Sarkar, Sumantra, "Leveraging Machine Learning to Combat AI-Driven Sophisticated Social Engineering Threats Online" (2024). *ICIS 2024 TREOS. 107. [SCOPUS]*

[5] Vo, T. A., Nguyen, L. A., **Do, B.**, Ma, C. T., Doan, S. T., & Ngo, S. H. (2025, December). TriFusion: GNN-Based Multimodal Fusion for 3D Object Detection in Autonomous Driving. In *International Symposium on Information and Communication Technology* (pp. 11-23). Singapore: Springer Nature Singapore.

[6] Ho-Vuong, K. L., Dang, N. H., **Do, B.**, & To, H. Q. (2026, August). ViCorpReviews: A Benchmark Dataset for Multi-Dimensional Sentiment and Hate Speech Detection in Vietnamese Workplace Context. In *2026 International Conference on Multimedia Analysis and Pattern Recognition (MAPR)* (pp. 192-197). IEEE.

JOURNAL PUBLICATION

[1] Le, B. H., Nguyen, H. H., Giang, T. T., Duong, C. H., **Do, B**., & Nguyen, P. N. D. (2026). Understanding social commerce adoption in Vietnam: A synergy of social, technological, and behavioral drivers. *Acta Psychologica, 266, 106925. [SJR-Q1, CiteScore-3.3, WoS-SSCI, IF-2.7, A-ABDC]*

LANGUAGE SKILLS

English TOEIC 670

Chinese TOCFL A1

REFERENCES

[1] Dr. Thi Tran (Email: ttran9@binghamton.edu)

Google Scholar: https://scholar.google.com/citations?user=L23zYkQAAAAJ&hl=en

Linkedin: https://www.linkedin.com/in/dr-thi-tran/

Assistant Professor of Management Information Systems, Binghamton University, State University of New York.

[2] Dr. Anh Dang (Email: ndang5@twu.edu)

Personal webpage: https://twu.edu/accounting-finance/faculty-staff/anh-dang-phd/

Department of Accounting & Finance, Texas Woman's University.